

AdvocateAuroraHealth

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Scope: ☐ AAH System ☐ ACH ☐ AAH IL Only ☐ AAH WI Only ☐ Site Only (Location Name): ☐ Department Only (Department Name): Health Informatics & Technology (HIT)		

I. PURPOSE

The purpose of the HIT Change Management process is to minimize service downtime by ensuring change requests are recorded and then:

- A. Reviewed in a controlled, standardized and consistent manner
- B. Evaluated for risk
 - 1. Technical perspective
 - 2. Customer perspective
 - 3. Of implementing or not implementing the change
- C. Authorized, prioritized, planned, tested, and implemented with the accompanying documentation to validate compliance with this standard
- D. If the unexpected occurs during the change, there shall be documented plans in place to restore service to the state it was before the change as quickly as possible

II. SCOPE

This Standard applies to Advocate Aurora Health (AAH) HIT team members and those team members and vendor and subcontractors whose changes fall under the oversight of HIT Service Management.

This document HIT Change Management Standard applies to all production services, applications, system assets (physical and virtual), network infrastructure assets, identified Pre-Production services, applications, system assets (physical and virtual), network infrastructure assets, all other supported HIT assets (physical or virtual).

- A. Out of Scope for Change Management:
Non-disruptive and non-impactful changes which do not:
 - 1. Alter or change data classification
 - 2. Modify the code of an application or device
 - 3. Modify the configuration or architecture of an application or service
 - 4. Alter the look and feel of an application

5. Alter team member workflows or processes
- B. The following are requests that do not go through the Change Management process:
 1. Printer/Workstation/Peripheral device installation and removal
 2. Routine team member access/security requests
 3. Modification of provider schedules
 4. Additions or changes to non-updating queries and reports

III. DEFINITIONS/ABBREVIATIONS

Change: The addition, modification, or removal of anything that puts at risk the delivery of a HIT service.

Change Management: The process responsible for controlling the lifecycle of all changes, enabling beneficial changes to be made with minimal disruption to HIT services and team members.

Configuration Management: The process responsible for managing and maintaining the enterprise Configuration Management Database (CMDB). Configuration Management tracks the lifecycle of all individual Configuration Items (CI) required to deliver a HIT Service

Incident: An unplanned interruption to an HIT service or reduction in the quality of HIT services. Failure of a configuration item that has not yet affected an HIT service is also an incident, i.e., failure of one disk in an array, or failure of a server in a cluster.

Incident Management: The process responsible for the managing the lifecycle of all incidents. Incident Management ensures that normal operation/function is restored as quickly as possible and the impact to patients, team members and other business operations is minimized.

Legally Restricted Confidential Information: includes any Advocate Aurora Information that is subject to laws or regulations related to its confidentiality, privacy, or security, and that may be disclosed only to certain identified individuals for which a record of disclosure may be maintained. Unauthorized disclosure of this information poses a significant risk and may result in criminal or civil fines, liabilities and breach notification obligations for Advocate Aurora

Practice: A way of working or a way in which work must be done. Practices include activities, processes, functions, standards and guidelines.

Problem: The cause of one or more incidents. Usually the cause is not known at the time the problem record is created, and the Problem Management process is responsible for further investigation: Cause Analysis.

Problem Management: The process responsible for managing the lifecycle of all problems. The output of Problem Management proactively prevents incidents from repeating or happening and minimizes the impact of incidents that cannot be prevented.

Request Management: The process responsible for managing the lifecycle of all Service Requests. Request Management ensures that Service Requests are fulfilled and delivered to the business as expected.

Subcontractor (also known as Third Party): An entity or person other than the vendor that offers to furnish goods or services of any kind under the contract of a vendor. **Vendor:** An external party providing paid goods or services to Advocate Aurora Health Care.

IV. STANDARDS

- A. AAH Three Standard Change Request Types
 - 1. Normal Change
 - a) Must adhere to specified lead times
 - b) Must be approved prior to implementing the change
 - 2. Emergency Change
 - a) Break/Fix (something is broken, change is needed to restore service)
 - b) Requires an associated incident or problem record
 - 3. Standard Change
 - a) Proven low risk/impact changes
 - b) Repeatable implementation plan
 - c) Pre-approved by leadership and Change Management
 - d) No lead time required
- B. AAH Three Risk Levels
 - 1. High
 - a) Significant impact or disruption (potential or actual) to the delivery of an HIT service
 - 2. Medium
 - a) Moderate impact or disruption (potential or actual) to the delivery of an HIT service
 - 3. Low
 - a) Zero to minimal impact or disruption (potential or actual) to the delivery of a HIT service
- C. AAH Approval Lead Times
 - 1. Normal High, Medium, and Low risk changes must meet their respective lead time, being in approved within the designated Service Management tool prior to implementation. See the reference document: "Lead Time Calendar for Planning Your Changes" (*available within the knowledge base of the designated Service Management tool*). *Normal changes that must be implemented which do not meet the required lead time will be considered "Expedited" requiring a business justification.*
 - 2. Each change request must be able to answer the following questions to ensure it is aligned with AAH's goals:
 - a) Who raised the change?
 - b) What is the reason for the change?
 - c) What is the return required of the change, its benefit?
 - d) What are the risks of implementing or not implementing the change? Risks can be positive or adverse.
 - e) What resources are required to build, test and implement the change? Resources, for example, are usually developers, system administrators or engineers.

- f) Who is responsible for the building, testing and implementing the change? Leadership team members are usually the responsible party(s).
 - g) What is the relationship with other changes – are there downstream / upstream dependencies?
- D. AAH Change Request Requirements
 - 1. Required Documentation

The following documentation is required for all Normal, Emergency, and Standard change types as part of creating and managing a change request. The detail required will vary, depending on the risk of the change, the greater the risk, the more detail is required.

 - a) A description of the change, written in a conversational style, minimizing or explaining jargon so anyone of the CAB can easily understand the change.
 - b) Impact to the Business
 - (1) A narrative of the impact to the business during the change itself in terms of performance, functionality, and availability.
 - (2) A narrative of impact to the business if the change does not go as planned in terms of performance, functionality, and availability.
 - c) Configuration Item(s)
 - d) Testing Documentation
 - (1) Pre-Implementation Testing Results
 - (2) Post-Implementation Validation Testing Plan
 - e) Implementation Plan
 - (1) Steps to implement the change. The plan includes any pre-work steps that need to be completed as part of the change, such as copying a file, backing up a database, etc.
 - f) Backout or Mitigation Plan
 - (1) Details of what actions/steps will be taken to recover from a change that did not go as planned.
 - g) Business Acceptance of the Change
 - (1) The business representative accepts the reason and scope of the change when there is direct impact to the business in terms of changes to the look and feel of the application, changes to team member processes and/or workflows, planned downtimes for implementation of the change, etc.
 - h) Team Member Communication - Team members affected by the change have been:
 - (1) Notified of the change.
 - (2) As needed, been trained on new functions / workflows / processes and procedures or modifications thereof.

- i) Incident or Problem Record Related to Emergency Change Requests
 - (1) As Emergency change requests are reserved for break/fix situations, a related incident or problem record within the designated Service Management tool must be identified on the change record.
- 2. Quality of Documentation

A change request and its accompanying documentation must stand on its own, meaning:

 - a) If the Assignee or implementer of the change would become unavailable to implement the change, a technical peer, relying on their expertise and the documentation in the change request, could flawlessly implement the change.
 - b) If a problem was encountered during the implementation, a technical peer could bring the application/service back to a steady, reliable state.
 - c) Statements must be validated with attachments or additional detail. For example, a statement regarding successful testing must be supported by attaching a document showing the results, or a narrative of what was tested noting pass/fail for each item.
- 3. Emergency Change Request Approval
 - a) All Emergency change requests must be in an Approved state in the designated Service Management tool within 6 calendar days of the Planned Start Date.
- 4. Change Request Closure
 - a) All completed change requests must be closed in the designated Service Management tool within a reasonable amount of time, prior to the beginning of the monthly HIT Change Management audit.
- E. Audit Criteria

Change requests and supporting documentation are maintained in the common service management tool used by all Advocate Aurora HIT team members.
- F. Disciplinary Action

Failure to comply with this standard may result in corrective disciplinary action up to and including termination of employment. Change Management will report compliance violations via communication to the affected team members' leader in accordance with the Change Management Standard Compliance Feedback procedure.

V. ROLES AND RESPONSIBILITIES

Role	Responsibility
A. Approver (IT Requester's Manager)	1. Ensures their team members follow the Change Management Standard, its supporting process and

Role	Responsibility
	procedures, to manage the lifecycle of change requests from their inception to their implementation and closure. 2. By approving the change request, the Approver infers they have: a) Reviewed the change request, ensuring all required documentation is part of the change request, either as an attachment, completion of a field or selection of data elements, such as Configuration Items, change requests and provide approval b) Approves the client notification c) Approves the risk and type of change 3. Monitor their groups change requests 4. Communicates events to their staff 5. Reviews and follows up with their staff on changes implemented as unsuccessful (backed out), and as deficient on the Monthly or Quarterly Change Request Audit. 6. Successfully completes required annual Change Management Training
B. IT Requester	The HIT resource who initially perceives the need for the change. Some examples of IT Requesters are: 1. An engineer replacing obsolete network hardware with newer-generation hardware with improved functionality 2. An engineer upgrading the capacity of a device 3. An engineer who needs to replace a defective part in a device 4. An application owner requesting a configuration and documentation change in response to a newly discovered vulnerability
C. Assignee	1. Ensures all required documentation is included in the change request 2. Create and maintains test plans/results 3. Performs or oversees pre-implementation testing as defined in the test plan 4. Performs or oversees the implementation of the change 5. Performs or oversees post-implementation testing as defined in test plan 6. Performs or oversees the backout/mitigation plan as defined in the change, if it is executed

Role	Responsibility
	7. Ensures workflow to indicate the change was implemented and the change request is closed. 8. Successfully completes required annual Change Management Training 9. Ensures all required documentation is included in the change request 10. Create and maintains test plans/results 11. Performs or oversees pre-implementation testing as defined in the test plan 12. Performs or oversees the implementation of the change 13. Performs or oversees post-implementation testing as defined in test plan 14. Performs or oversees the backout/mitigation plan as defined in the change, if it is executed 15. Ensures workflow to indicate the change was implemented and the change request is closed. 16. Successfully completes required annual Change Management Training
D. Change Advisory Board (CAB)	Within Advocate Aurora HIT, the Change Advisory Board (CAB) is an authoritative and representative group of HIT team members who are responsible for reviewing, assessing and approving High and Medium risk change requests. Representatives include team members from Applications, HIT Operations, Data Center Operations, Security, and Technical Engineering, and others as designated. At this time, CAB members are expected to understand and represent: 1. The expertise of their technical domain (application or infrastructure) 2. Their technical domain (application or infrastructure) change request(s) and its impact, explaining in terms easily understood by CAB members outside of their domain 3. How their change impacts other domains (application or infrastructure) 4. The business risk and impact to team members affected by the change from the view point of those team members 5. The risk and impact to patients affected by the change from the view point of those patients

Role	Responsibility
	6. Third parties, making changes to their environments (application or infrastructure) external to, or on the grounds of Advocate Aurora, that affect Advocate Aurora's digital ecosystem 7. Review of changes implemented as unsuccessful 8. Potential risks related to privacy, security, and compliance. To the extent there are privacy related risks or concerns related to Legally Restricted Confidential Information, which includes Personally Identifiable Information, Protected Health Information and/or payment card industry data, with a particular change, the HIT Security team will escalate to the Chief Privacy Officer and/or their delegates Whether the meeting is in-person or virtual, primary CAB members are expected to participate in the review and approval of all change requests on the agenda for review.
E. Affected User	The Affected User is the primary team member whom the change will impact. Some examples of Affected Users are: 1. A Patient Service Market leader desiring a new or changed feature on an application or service 2. A Hospital Administrator desiring a new or changed feature on an application or service 3. An application or service user desiring a new or changed feature on an application or service
F. Change Process Owner	1. Has the overall responsibility for ensuring the suitability of the Change Management process to the Advocate Aurora HIT organization. This is fulfilled by: a) Sponsoring the process by ensuring the Change Manager has adequate resources to design the Change Management process conforming to best practices and meeting the needs of the organization. b) Governing authority for the process and addresses performance or changing business needs.

Role	Responsibility
	c) Arbitrating in any dispute over the allocation of responsibilities and to sponsor the communication campaign to promote awareness and acceptance of the Change Management process. d) Providing the description, mission statement, process objectives and metrics to measure success and obtain formal approval for the Change Management process and its associated procedures. e) Monitoring and reviewing the execution of the Change Management process at a high-level, ensuring it remains consistent with the organization's current culture and ITSM strategy and ensure coordination with all other ITSM and HIT Infrastructure Management processes. 2. Is responsible for the execution of the Change Management process. This includes operating the defined and agreed process, ensuring it interfaces with all other relevant processes, setting targets and reviewing the effectiveness and efficiency of the process, performing process audits and managing the process improvement cycle. a) Develop and maintain the process b) Review the effectiveness and efficiency of the process c) Produce management reports d) Coordinate Change Management staff e) Allocate resources for support 3. Successfully completes annual Change Management Training
G. Change Manager	1. Facilitates/leads CAB (Change Advisory Board Meetings) 2. Maintains the change schedule and projected service outages 3. Authorizes standard change templates/models
H. Change Management Analyst	Assists the Change Manager in administration of the Change Management process by: 1. Reviewing pre-implemented change requests, with a focus on higher risk changes

Role	Responsibility
	a) Change was recorded correctly (including required documentation, required fields, required data) b) Change is the appropriate Type c) Change meets its Lead Time d) Change has appropriate Risk Level 2. Managing and conducting periodic change review meetings 3. Audit post-implemented change to ensure: a) Change was recorded correctly (including required documentation, required fields, required data) b) Change is the appropriate Type c) Change meets its Lead Time d) Change has appropriate Risk Level e) Change was implemented f) Change was appropriately closed out g) Maintain Change Management supporting documentation, including, but not limited to: (1) Standard Process (2) Create and manage a change request (3) Approval of a change request 4. Managing change postmortems a) Creating and compiling Change Management metrics b) Contribute to process development and improvement c) Create and maintain training courses and reference materials d) Reviewing emergency changes, both from a business perspective and from a technical standpoint, making thoughtful, appropriate decisions e) Ensure all required documentation for the Emergency change is either recorded in fields embedded in the change request itself, or attached to the change request 5. Successfully completes annual Change Management Training
I. Implementer(s)	1. Completes tasks associated with approved change requests

VI. CROSS REFERENCES

A. Other References (located in the Knowledge base of the designated Service Management tool):	1. Approver Cheat Sheet 2. Change Management CAB Guidelines and Member Expectations 3. Expedited Change Requests 4. Reviewing and Approving change requests 5. Assignee Cheat Sheet 6. Change Management Standard Compliance Feedback Procedure 7. Change Management Standard Compliance Feedback Form 8. Creating a Change request 101 9. Emergency Change Workflow Tasks 10. Lead Time Calendar for Planning your Changes 11. Normal Change request Workflow Tasks 12. Change Management Quick reference 13. Risk Assessment Cheat Sheet 14. Adding Change Tasks and Implementers 15. Change Request Maintenance and Restricted Change Windows 16. Requesting a Standard Change Request Template 17. HIT Change Management Audit Process <i>(all reference documents are available within the Knowledge base of the designated Service Management tool)</i>
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VII. RESOURCES AND REFERENCES

A. Supported Compliance Controls:	1. HIPAA a) §164.316(a) b) §164.316(b)(2)(iii) 2. HITRUST a) Change Management b) Segregation of Duties c) Separation of Development, Test, and Operational Environments d) Control of Operational Software e) Change Control Procedure 3. AAH Information Classification 4. Informatics Technology Infrastructure Library (ITIL): A set of detailed practices for IT service management (ITSM) that focuses on aligning IT services with the needs of business. 5. NIST Cybersecurity Framework: PR.IP3 Configuration change control processes are in place
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	6. PCI DSS 3.2 a) Follow change control processes and procedures for all changes to system components. b) Separate development/test environments from production environments and enforce the separation with access controls. c) Separation of duties between development/test and production environments d) Production data (live PANs) are not used for testing or development e) Removal of test data and accounts from system components before the system becomes active / goes into production. f) Change control procedures must include the following: g) Documentation of impact. h) Documented change approval by authorized parties. i) Functionality testing to verify that the change does not adversely impact the security of the system. j) Back-out procedures. k) Upon completion of a significant change, all relevant PCI DSS requirements must be implemented on all new or changed systems and networks, and documentation updated as applicable.
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VIII. ATTACHMENTS

Not Applicable

Version History

Version	Author	Changes	Date of Approval
1.0	Jo Bruns	Original	09/30/2014
2.0	Steve Sonderman	Revision	12/01/2015
3.0	Steve Sonderman	Revision	12/17/2015
4.0	Steve Sonderman	Removed Emergency Change Advisory Board	03/31/2016
5.0	Steve Sonderman	Definition changes, risk levels, Standard Requirements, Role	09/27/2016

Version	Author	Changes	Date of Approval
		Responsibilities, and Service Owner Responsibilities	
6.0	Steve Sonderman	Inclusion of Operational Readiness component, addition of Emergency Change Approval time and update to Closure Time in Standard Requirements section, addition of training requirement for Change Manager and Change Management Analyst roles, and updates to Standards References section	N/A
7.0	Steve Sonderman / Diana Waniorek	Added additional text under the "Standard" section to further define when a change requests does and does not need to be created	N/A
8.0	Steve Sonderman / Diana Waniorek	Updated verbiage to reflect organizational changes for Advocate Aurora Health. Made multiple updates to align with the update change processes for the new ITSM tool (ServiceNow): Removed references for Operational Readiness. Removed reference to Priorities and second approver. Added roles for Change Manager and Requester.	N/A
8.1	Steve Sonderman / Diana Waniorek	Updated verbiage and included definition for Legally Restricted Confidential Information	N/A
8.2	Steve Sonderman / Diana Waniorek	Updated to remove specific days requirement for change request closure.	N/A
9.0	Steve Sonderman / Diana Waniorek	Updated to new format for PolicyTech, removed extraneous text no longer needed in the Standard.	N/A